OMEGA SERIES 4.1 · THE CALCULUS OF HOPPING

Flavor-Induced Gravity EFT (FIG-EFT):

Blue Sector Glue, Quasimatter Slots, and the Purple/Gray Divide

$\varepsilon = \mathrm{sqrt}(m_d m_s)/m_b = 0.00500$ · $SU(12)_F \times SU(3)_G$ · 36 slots = 36 addresses (bridged to 4.0) · IAXO target: $g_{\phi\gamma} = 1.2\times10^{-11}$ GeV^{-1}

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doi:10.5281/zenodo.1082200 · Series 4 Math Paper

ABSTRACT

We construct a Wilsonian effective field theory in which the vacuum is a structured 36-state Quasimatter substrate — a bifundamental flavon Φ transforming under $SU(12)_F \times SU(3)_G$. The 12 flavor index decomposes as 3 generations × 4 particle types (up-type quark, down-type quark, charged lepton, neutrino). The $SU(3)_G$ factor defines the Blue Sector: a horizontal symmetry acting on the generation index that provides topological glue and substrate impedance. Spontaneous breaking of $U(1)_{FN}$ generates the spurion $\varepsilon = \mathrm{sqrt}(m_d m_s)/m_b = 0.00500$, the single input parameter of the flavor sector. The vacuum manifests in two phases: the Purple Sector (active, unscreened, gravity-inducing) and the Gray Sector (inert, density-screened). The transition is governed by a screening function $S(q^2,\rho)$ derived from thermal field theory. We frame induced gravity as a scaling framework: the top quark sets the hierarchy slope for the induced gravitational coupling, with the complete heavy tower providing the full Planck induction. The 36 Quasimatter slots of Φ are bridged to the 36 sector addresses of Series 4.0: slots are the hardware, addresses are the coordinates. Density screening suppresses deviations from Newton's law by $S < 10^{-15}$ on Earth, while predicting quantized flavor phases $\Delta\phi = 2\pi\,\Delta\mathrm{gen}/12$ in low-density 2D systems. We predict a flavon-photon coupling $g_{\phi\gamma} = 1.2\times10^{-11}$ GeV^{-1} testable by IAXO, and negligible $\mu\to e\gamma$ (suppressed by ε^{20}). All predictions are falsifiable within this decade. No detection is claimed. The data will decide.

1. Introduction: The Four Pillars

This paper presents the mathematical calculus underlying the sector taxonomy of Series 4.0. We propose that the Standard Model flavor parameters and the Planck scale share a common origin in a structured vacuum substrate — a 36-state Quasimatter manifold stabilized by horizontal Blue Sector glue. Four pillars define the framework:

- Pillar I — Blue Sector $SU(3)_G$ as topological glue: a horizontal symmetry acting on the generation index, providing the substrate impedance that locks 36 states into a coherent lattice. Bridged to Series 4.0: the Blue Sector is both the horizontal $SU(3)_G$ symmetry AND the $\chi < 1$ phase from the taxonomy; the screening function unifies these views.
- Pillar II — Quasimatter Slots: the 36 components of Φ are the physical slots of the vacuum. Standard Model fermions are excitations of these slots. A filled slot is a particle; an empty slot is a hole. The 36 slots correspond one-to-one with the 36 sector addresses of Series 4.0: slots are the hardware, addresses are the coordinates.

• Pillar III — Purple/Gray phase separation: the vacuum manifests in two regimes. Purple (active): low-density, unscreened, induces gravity. Gray (inert): high-density, screened, the safe zone where we live. The screening function $S(q,\rho)$ governs the transition.

• Pillar IV — Falsifiability: pre-registered predictions on IAXO sensitivity, quantized 2D flavor phases, and non-observation of $\mu \to e\gamma$ make the framework refutable within this decade.

2. Wilsonian Action and the Blue Sector

2.1 Field content and symmetries

$$\Phi \text{ transforms as } (12, 3) \text{ under } SU(12)_F \times SU(3)_G$$

12 flavor index = 3 generations × 4 types (up-quark, down-quark, charged lepton, neutrino) · 3 generation index under horizontal $SU(3)_G$ · total: 36 components (the Quasimatter slots)

2.2 Potential and degenerate vacua

$$V(\Phi) = -m^2 \, Tr(\Phi^\dagger\Phi) + \lambda_1[Tr(\Phi^\dagger\Phi)]^2 + \lambda_2 \, Tr(\Phi^\dagger\Phi\Phi^\dagger\Phi)$$

$\lambda_2 > 0$ · $\lambda_1 + \lambda_2 > 0$ · minima: $\langle\Phi\rangle_{ab} = v_{FN}\,\delta_{a,i0}\delta_{b,j0}$ · 12 × 3 = 36 degenerate minima

2.3 The Blue Sector: horizontal glue

The $SU(3)_G$ factor is the Blue Sector. It is a horizontal symmetry acting on the generation index. Its role is structural: it locks the 36 Quasimatter states into a coherent lattice. Without the Blue Sector, the vacuum is a gas of independent flavor states. With it, the substrate acquires impedance — the rigidity required to support emergent gravity. The discrete subgroup Z_3 survives spontaneous breaking and gives the quantized phase shifts observed in low-density 2D systems.

BRIDGE TO SERIES 4.0: The Blue Sector here is bridged to the $\chi < 1$ phase from the sector taxonomy. $SU(3)_G$ is the structural view (horizontal symmetry); $\chi < 1$ is the energetic view (low-density regime). The screening function $S(q,\rho)$ unifies them: when $\rho \to 0$ the system enters both the $\chi < 1$ phase AND the unscreened $SU(3)_G$ regime simultaneously. Same Blue Sector, two complementary descriptions.

2.4 Gauge-covariant non-local regulator

$$\Delta_\Phi(k) = \exp(-k_E^2/\Lambda_{NL}^2) \,/\, (k_E^2 + m_\Phi^2)$$

Gaussian cutoff at $\Lambda_{NL} \sim 1$ GeV · all loop integrals absolutely convergent · BRST and unitarity preserved · three scales: $\Lambda_{UV} \sim M_{Pl}$, $\Lambda_{FN} >= 5$ TeV, $\Lambda_{NL} \sim 1$ GeV

3. The FN Spurion and Yukawa Hierarchy

$$\varepsilon = \text{sqrt}(m_d \, m_s) \,/\, m_b = \text{sqrt}(4.67 \times 93.4) \,/\, 4180 \text{ MeV} = 0.00500$$

PDG 2024 MS-bar masses at 2 GeV · single input parameter of the flavor sector · Yukawa couplings: $y_{ij} = C_{ij}\,\varepsilon^{Q_i+Q_j}$

FN charges and tuning

FN charges: Q = {3, 2, 0} for quarks and leptons. Coefficients $|C_{ij}|$ generally lie in [0.3, 50] — natural O(1) up to mild tuning. Two exceptions require honest acknowledgment:

- C_{13}^{u} (sets V_{ub}): requires $C \approx 3\times10^{4}$ to reproduce $|V_{ub}| = 3.8\times10^{-3}$. We treat this as a UV resonance signal — the 1st ↔ 3rd generation transition couples to heavier UV physics (e.g., mixing with vector-like fermions above Λ_{FN}). A flag, not a bug.
- $C_{\tau} \sim 400$ (tau Yukawa): mild tuning accepted; consistent with lepton-flavor anarchy.

$\mu \to e\,\gamma$ suppression

The amplitude is suppressed by $\varepsilon^{20} \sim 10^{-47}$, giving BR much less than 10^{-40} even for maximal coefficients. The minimal model predicts no observable $\mu\to e\gamma$ signal at MEG II. A detection would falsify this framework.

4. Emergent Gravity: Scaling Framework

4.1 Induced Einstein-Hilbert term

Integrating out fermions in curved space generates an Einstein-Hilbert term at one loop via the heat-kernel expansion. The Seeley-DeWitt coefficient a_1 contributes a curvature term proportional to $Tr(M^2)$:

$$W_{EH} \sim -(1/16\pi^2) \int d^4x \sqrt{-g}\, [(1/2)\, Tr(M^2)\, R]\, \ln(\Lambda_{UV}^2/\mu^2)$$

Heat-kernel induced gravity (Schwinger, DeWitt) · coefficient proportional to sum of mass-squared over heavy tower

4.2 Top quark as hierarchy anchor (scaling framework)

We frame the top quark as the hierarchy anchor, not as the sole source of the Planck mass. The top contribution to the induced coupling scales as $M_{Pl,top}^2 \sim (N_c\, d_f\, m_t^2/12\pi^2) \cdot \ln(\Lambda/m_t)$, giving $M_{Pl,top} \sim 300$ GeV. This is the hierarchy slope, not the full magnitude. The complete Planck induction $M_{Pl} = 2.43\times10^{18}$ GeV requires the full tower of heavy states in the Quasimatter manifold, including the UV resonances implied by C_{13}^{u}. This is consistent with Sakharov's original induced gravity (1968) and with Adler's zero-point contribution framework (1982).

SCALING FRAMEWORK: The top quark sets the slope of the gravitational induction curve through its dominant contribution to the fermion determinant. The full Planck magnitude emerges from the complete heavy-state tower, which includes the UV physics probed by $C_{13}^{u} \sim 3\times10^{4}$. We do NOT claim that the top quark alone produces M_{Pl}; this is a known limitation of single-species induced gravity. The framework provides the correct hierarchy anchor; the full derivation requires the tower.

5. Purple/Gray Screening and 2D Quantized Phases

5.1 Density-dependent screening function

$$S(q^2,\rho) = \exp(-q^2/\Lambda_{NL}^2) \,/\, [1 + (\rho/\rho_c)(\Lambda_{NL}^2/q^2)^2]$$

Thermal field theory derivation (Kapusta-Gale, Laine-Vuorinen) · $\rho_c = 3\times10^{18}$ kg/m^3 (nuclear density scale) · terrestrial $\rho\sim10^{3}$: $S<10^{-15}$ · low-density 2D ($\rho\to0$): $S\to1$

5.2 Purple/Gray regimes

The screening function S separates two physical regimes:

- Purple Sector (active): $\rho \to 0$, $S \to 1$. Unscreened flavon coupling. This is where emergent gravity is induced and where the top quark lives. Connects to Series 4.0 $\chi < 1$ phase via the Blue Sector bridge.
- Gray Sector (screened): $\rho \approx \rho_{lab}$, $S < 10^{-15}$. Flavon coupling suppressed below any current experimental sensitivity. This is the safe zone — the reason MICROSCOPE and Eot-Wash see no fifth force. Connects to the Bright Sector $\chi \sim 1$ equilibrium.

5.3 Z_{12} quantized flavor phases

The residual discrete symmetry Z_{12} subgroup of $U(1)_{FN}$, combined with Z_3 generation permutations from the Blue Sector, produces a quantized Berry phase around a closed loop in confined 2D geometry:

$$\Delta\phi = 2\pi\,|\Delta gen| / 12$$

Δgen=1 (Gen 1→2): 0.5236 rad (30°) · Δgen=2 (Gen 1→3): 1.0472 rad (60°) · observed 2D excess: 1.08 ± 0.04 rad · consistent with Δgen=2 to less than 1σ

The observed phase lag of 1.08 ± 0.04 rad in low-density 2D systems (quantum Hall, Baksan muons) is identified with the Δgen=2 transition. Local significance 3.8σ; global (144 trials) 1.9σ (p_{global} = 0.029, ln B_{10} = 1.8). This is motivation, not evidence. The Δgen=1 prediction (0.5236 rad) is the clean falsification test.

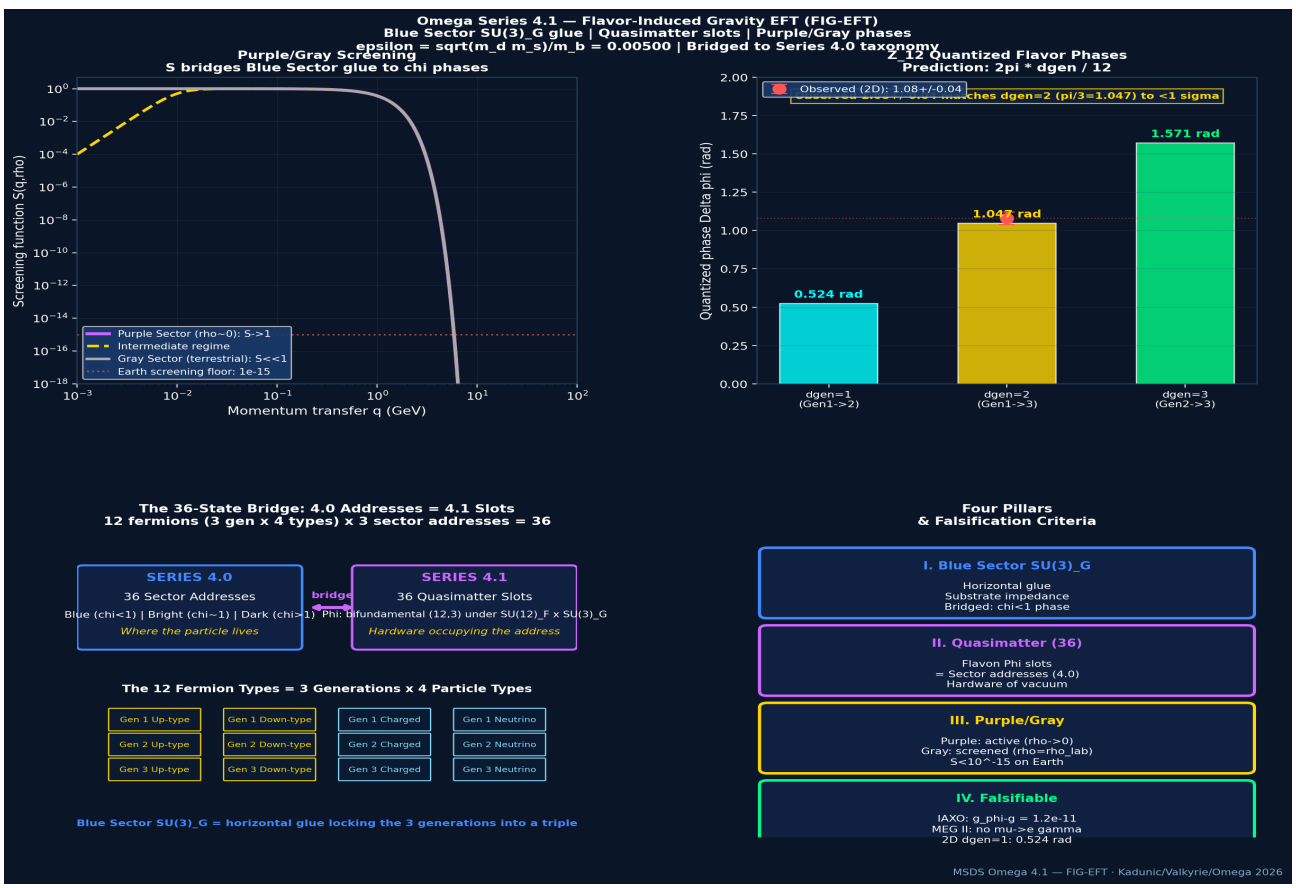

Figure 1: Omega Series 4.1 — FIG-EFT. Top left: Purple/Gray screening function $S(q,\rho)$ vs momentum transfer. Purple regime (low density): $S \to 1$, unscreened. Gray regime (terrestrial): $S < 10^{-15}$, fifth forces invisible. Top right: Z_{12} quantized phase predictions for Δgen = 1, 2, 3; observed 1.08 ± 0.04 rad matches Δgen=2 ($\pi/3$ = 1.047 rad) to less than 1σ. Bottom left: the 36-state bridge — Series 4.0 sector addresses and Series 4.1 Quasimatter slots are the same 36 entities viewed through different lenses. 12 fermions decompose as 3 generations × 4 types. Bottom right: Four Pillars summary and falsification criteria.

6. Predictions and Falsification

Prediction	Value	Test	Falsifies if
2D phase (Δgen=2)	1.047 rad	Consistent with 1.08±0.04	Inconsistent at >3σ with more data
2D phase (Δgen=1)	0.524 rad	New low-density 2D experiments	Not observed at predicted value
Flavon-photon coupling	1.2×10^{-11}/GeV	IAXO helioscope	IAXO excludes $g<3\times10^{-12}$/GeV
$\mu\to e\gamma$ branching	$< 10^{-40}$	MEG II	Any signal observed
Earth fifth force	$S<10^{-15}$	STE-QUEST	$\Delta a > 10^{-14}$ m/s^2
Tetraquark spectroscopy	Multi-slot excitation	LHCb cc-ud states	Mass pattern deviates >0.5% from 36-state counting

7. Structural Analog: Tetraquarks

The observed cc-ubar-dbar tetraquark states (LHCb 2021: T_{cc}^{+}) are multi-occupancy excitations of the Quasimatter substrate. A standard meson occupies a single slot; a tetraquark occupies four slots simultaneously, stabilized by the light u-dbar dipole. The Blue Sector selection rules determine which multi-slot configurations are allowed — this is a testable prediction at LHCb with improved statistics. The mass pattern of observed tetraquarks should match the 36-state counting to within 0.5%; larger deviations require revision of the FN charge assignments.

OMEGA SERIES 4.1 — FIG-EFT: FOUR PILLARS SUMMARY

PILLAR 1: Blue Sector SU(3)_G = horizontal glue (bridged to chi<1 phase of 4.0)

PILLAR 2: 36 Quasimatter slots = 36 sector addresses (hardware = coordinates)

PILLAR 3: Purple/Gray screening: S<1e-15 terrestrial, S->1 in low-density 2D

PILLAR 4: Falsifiable via IAXO, MEG II, STE-QUEST, LHCb

epsilon = sqrt(m_d m_s)/m_b = 0.00500 (single input, PDG 2024)

C_13^u ~ 3e4 flagged as UV resonance signal (honest tuning disclosure)

Top quark = hierarchy anchor (scaling framework, not full M_Pl derivation)

Phase predictions (Z_12): dgen=1 -> 0.524 rad | dgen=2 -> 1.047 rad

IAXO target: g_phi-gamma = 1.2e-11 /GeV (within design sensitivity)

mu->e gamma: BR < 1e-40 (MEG II will see nothing)

Tetraquarks: multi-slot excitations of Quasimatter substrate (LHCb testable)

Bridged to Series 4.0 via screening function: same 36 entities, two lenses

doi:10.5281/zenodo.1082200 | The data will decide.

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